

A lean and mean strategy: a data migration industrial study

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SUMMARY

Many companies have legacy assets to be integrated or modernized. Like today's software developments, migration projects are faced with steadily increasing demands for efficiency: migration has to be carried out faster, better, and cheaper. At the same time, migration complexity increases, and migration projects remain challenging despite the companies' extensive experience.

To cope with such efficiency demands, companies need a well-defined migration strategy. In previous work, we have defined such a migration strategy called lean and mean strategy. In this paper, following technical action research, we apply our migration strategy, and evaluate it, in two large industrial case studies in the field of data migration. The use of the strategy in our case studies reveals the benefits of lean and mean strategy for meeting the efficiency demands. Copyright © 2013 John Wiley & Sons, Ltd.

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1. INTRODUCTION

Many companies migrate their legacy assets to modern paradigms such as service orientation. This happens frequently as a result of mergers and reorganizations. Think of migration examples like the ABN-AMRO/Fortis banks, KLM/Air France, or Océ/Canon mergers, involving thousands of applications in their information technology (IT) portfolio and the data of millions of customers. These migrations typically have to deal with many challenging problems. Those problems do often relate to the ways the new, post-merger IT assets are envisioned, compared to the ways the preexisting assets are already developed [1–3]. For instance, the new IT assets are envisioned to be a set of small pluggable software elements (e.g., services), whereas the preexisting assets are large, monolithic systems; the new assets need to be designed under open world assumption and be distributed across organization boundaries, whereas the preexisting assets are often owned and controlled by a single owner. Not to mention data model inconsistency, functional overlap, and architectural and platform inconsistency between the new and preexisting assets. Moreover, in carrying out migration projects, companies have to address various demands for high quality, fast delivery, and decreasing costs. Hence, migration projects remain challenging despite the companies' extensive experience.

Logica, the consultancy company involved in this study, had the problem of achieving efficiency in their migration projects. One of the main roots of this problem, emphasized by managers of the business migration sector, was inadequate reuse of past experiences. In this regard, a manager said: 'essentially there are many similarities in migration projects, but usually when a new project arrives we start all over again. We need to avoid repeating what we already know'. As noted by the managers, reusing the existing assets can reduce time and cost while ensuring proven quality. This is, however, not easily achieved in the presence of many different complexities such as incompatibility between data models that lead to inefficiency.

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This paper addresses the efficiency problem of Logica by applying a migration strategy that has two features: (i) reusing past experience; and (ii) employing a light-weight migration approach. Our first solution for achieving efficiency is to reuse past experience. Consultancy companies are faced with similar migration projects and therefore have vast experience in migration. As shown by various works [4, 5], reusing experience in the form of activities or knowledge is essential for efficient and cost-effective migration. To be able to reuse experience, it needs to be explicitly captured. Capturing the experience can be achieved by a general model for migration strategies that enables reuse by identifying the fundamental commonalities between migration projects.

Our second solution to address demands for efficiency centers around light-weight migration. Light-weight software development approaches have been adopted by many companies over the past decade [6]. Although some claim that without a complex, concise, and rich software process, efficiency cannot be achieved [7–9], many studies have shown that light-weight approaches are in-line with the team's way of working, focus on most important tasks, and lead to efficiency [5, 10, 11].

In previous work [12], we defined a strategy that generalizes the practice of industrial service-oriented architecture (SOA) migration into a lean and mean migration strategy. Inspired by having the both features mentioned earlier, that is, (i) supporting reuse of the common practice; and (ii) being light-weight, this work proposes such strategy as a solution for the efficiency problem of Logica. The basic idea behind the lean and mean strategy is to separate the fundamental knowledge and activities common among any migration project from the project-specific ones. Following this idea, we develop an empirically grounded lean and mean strategy and evaluated it in two case studies in the field of data migration. Data migration, according to Gartner [13], is the most important problem in legacy migration, whereas failed data migration efforts are common. By acknowledging that most migration projects will require migration of data, managers of Logica proposed data migration as the focus of this study.

In this study, we serve two goals: (i) Logica's goal to improve their efficiency; and (ii) our research goal to improve the relevance of our lean and mean strategy by trying it out in two large industrial case studies. To achieve these goals, we follow technical action research (TAR) [14]. Related to our first goal, we develop the lean and mean strategy for data migration (Section 4). In addition, the uses of lean and mean strategy reveal the following benefits each related to efficiency: (i) increased reuse; (ii) improved problem-solving; (iii) capturing and supporting existing-way-of-working (Section 5). With respect to our second goal, the industrial perspective of this study brings insight about the generalizability of our lean and mean strategy (Section 6). Moreover, interesting observations and lessons learned further contribute to the novelty of our approach (Section 9). It is worth mentioning that although data migration constitutes the subject of our case studies, the technical aspects of data migration are not the main focus of this work. Rather, our focus is on providing a common ground to consolidate and capture knowledge and experience according to company-wide practices and policies.

This paper is organized as follows. We introduce the notion of lean and mean strategies in Section 2. Section 3 introduces study design. In Section 4, we present the lean and mean strategy for data migration. Then, Section 5 explains how such a strategy facilitates efficiency. Section 6 presents the generalizability of such an approach. Section 7 discusses the related work. In Section 8, we discuss threats to validity. Section 9 presents our additional observations followed by conclusion of our paper.

1.1. Industrial case studies

The case studies in this work are in the area of data migration. It is recognized that data migration is an important and usually overlooked aspect of migration or generally reengineering methods [13]. Migration of data plays an integral role in many migration scenarios in which different types of assets, ranging from legacy code to business processes, are ported to a new environment. This suggests that different types of migration strategies (e.g., strategies for migration to services, or cloud) need to especially care for the migration of the data, too. For instance, in the context of SOA migration, the following three scenarios, emerged from our collaboration with industrial practice [15], exemplify the integral role of data migration when migrating to services.

- In many SOA migration projects, the goal is to reengineer the legacy systems into a set of services which can be dynamically selected and are distributed across organization boundaries. Such a migration introduces a shift in the ownership: from legacy systems that are owned and controlled

by a single organization to a set of services that are owned by multiple parties. This shift in ownership is not only reflected on systems, but also on the data managed by those systems. Consequently, to address such a shift, the data needs to be reengineered too.

- When enterprises want to improve their business process and integrate multiple, redundant, and fragmented systems, they need to consolidate the associated data structures. This usually requires reengineering of data.
- When the goal is to move to new delivery solutions such as software-as-a-service, many existing assets need be reengineered. Data is one of the most critical ones as new liability or auditing issues may arise because sensible data are no longer stored exclusively on premise.

In all these scenarios, the migration strategy needs to support understanding, transforming, and porting the data to the new SOA environment.

1.1.1. Case descriptions. This work includes two case studies, referred to as Case I and Case II. Case I was a post-mortem analysis of a data migration project, whereas Case II was studied while it was still active. The reason behind selecting one post-mortem and one ongoing project lies in our proposed plan for treating the efficiency problem of Logica. In short, the treatment plan constituted first recovering the migration strategy of the post-mortem project (i.e., Case I) and represents it in form of lean and mean. Next, for defining the migration strategy of the ongoing project (i.e., Case II) the plan was to take the Case I as a reference and build the Case II strategy based on such a reference. In the following, we briefly introduce the two case studies. Table I provides an overview of the case studies. More details about these two cases and the adopted migration strategies follow in the remainder of this paper.

Case I Our first case study (Case I) was a successfully completed data migration project addressing the legal merger of two large banks. These two banks were merged with the main goal of creating synergies. Logica, together with another consultancy company, was in charge of the data migration of their nearly 1.6 million retail banking customers. In early stages of this merger, it was decided that the systems and infrastructure of the acquired bank will be retired. This required the data of the customer to be moved to the new merged bank.

One of the key distinguishing factors of this project was the high volume of data (in the order of 100 terabytes). In addition, there were two major complexities in this data migration: (i) there were various undocumented, and complex relationships between business entities. Those relationships were stored, in form of business rules, in the databases in which they were mixed with data; and (ii) complex interdependencies between 20 applications. In addition, in this project, there were strict time constraints. For instance, the data had to be loaded only in weekends. The migration approach adopted was incremental and fully automated. The migration was partitioned into 16 increments. Each increment included the validation and load of 100 000 customers, their products, and financial information.

Case II The second case study (Case II) was an ongoing data migration project of Logica. Similar to Case I, data migration was because of the merger of the private banking sections of two banks. In this project, Logica was in charge of full migration of more than 140 000 private banking customers' data.

One of the distinguishing factors in this project was the demand for high-level of business involvement. To maximize customer's satisfaction, the business stakeholders (e.g., account managers) needed to ensure flexibility offered in private banking products. This implies that the data mappings and data conversions had to embed such flexibility as well. In addition, the following complexities made this project specially challenging: (i) strict business constraints; and (ii) high level of business involvement. With regards to strict business constraints (e.g., regulations, trust, and ownership), for instance, only the product manager dedicated to each customer was allowed to access customer's data. This made the role of product managers, as trusted parties, important in negotiations required for data mappings. Related to high level of business involvement, this project had many stakeholder types. Considering the stakeholder's high-level of involvement in the data migration, it was specially challenging to carry out certain activities such as data mapping.

Table I. Overview of case studies.

Case study	Data volume	Complexity	Constraints	Migration approach	Size of Logica team
Case I	In the order of 100 terabytes	1-business rules mixed with data 2-complex inter-dependencies between 20 applications	1-time window of weekends for data load	Incremental, fully-automated	30
Case II	In the order of 100 gigabytes (20% for customers and products and 80% for transactional history)	1-strict business constraints, 2-high level of business involvement, 3-complex interdependencies between 25 applications	1-time window of weekends for data load, 2-data can be accessed by account manager only, 3-the data of VIP customers needs to be specially confidential and cannot be in the same cluster as other customers, 5-business stakeholders need to have direct influence on data transformation	Big-bang, semi-automated	100

2. CONTEXT AND GROUNDWORK

With the purpose of increasing efficiency in mind, we proposed Logica the idea of capturing past experience in a reference model, which would indicate the fundamental elements repeating in every migration project. This idea led us to apply the notion of lean and mean strategies to Logica. In the following texts, we first explain our definition of efficiency for data migration projects. Next, we describe the basic concepts of the lean and mean strategy, followed by our proposed treatment for Logica.

2.1. Efficiency in migration projects

There is no universally accepted definition of efficiency in data migration. In this work, we define efficiency as reducing the total life-cycle cost and adding value for the customer, the product, and the enterprise. This definition embraces the following two aspects: (i) reducing cost and time to market; and (ii) providing value to the customer or the enterprise.

Although the first aspect of this definition—reducing cost and time to market—is in-line with the classic definition of efficiency in software engineering [16], the second aspect—providing value to the customer or the enterprise—is in-line with the definition of effectiveness [17]: the capability of the migration approach to add value for the customer and the product. Considering that our definition addresses both aspects, in this work, the terms time-to-market, cost, and effectiveness fall under the umbrella of efficiency.

2.2. Background on lean and mean migration strategy

The lean and mean strategy centers around the idea of separating the context-generic elements of migration from the context-specific ones. In any migration approach, there exist elements that are common for all migration efforts, independent from the context or the project at hand. We call those elements the *core elements* of migration. Although, the core elements, in one way or another, appear in all migration approaches, they are usually blended with context-specific details. The lean and mean strategy separates the core elements from the more specific ones called the *extensions*. Migration strategies that are formed in this manner are both lean (screening out details specific to the project at hand) and mean (covering specific needs of the project).

Our treatment for Logica was based on a lean and mean migration strategy that was developed in our earlier work [12], hereafter referred to as lean and mean-G (where ‘G’ stands for ‘Generic’). With the goal of identifying the industrial migration approaches, we have carried out an industrial interview survey in seven SOA solution provider companies, followed by a panel of experts [12]. Once we got beyond the great variance in terminology and description of these approaches, we found that they have a lot in common. More precisely, the approaches shared similar set of activities and similar set of input/output knowledge captured in the lean and mean-G. Although the core made of both knowledge and activities in Figure 1 generalizes the practice of industrial migration, extensions are needed whenever specialized uses are needed. The meta-model presented in Figure 2 represents the constructs of the lean and mean strategy and their relationships. In the following, we further explain these constructs in the core and extensions. We point out to the constructs of the meta-model in typewriter (e.g., core Activity or small core knowledge).

Core The core includes the essential elements that are common in every migration approach. The core is lightweight and simple and reflects the following two main aspects of the migration strategy ‘what is the input/output knowledge’ (core knowledge) and ‘what has to be done’ (core activities). The core is represented using core views including the two different, yet interrelated, views of core knowledge and core activity. Figure 1 represents an example of the two-view representation of the core for migration to services. core knowledge is (i) a set of stripped-down knowledge elements that shape and drive the migration (i.e., the 15 knowledge elements shown in Figure 1); and (ii) a set of knowledge conversions. Core knowledge can be customized for each migration project. This customization includes refinement of the knowledge elements or addition of some more input/output knowledge elements. Section 4.1.2 provides an example of the customization of knowledge view.

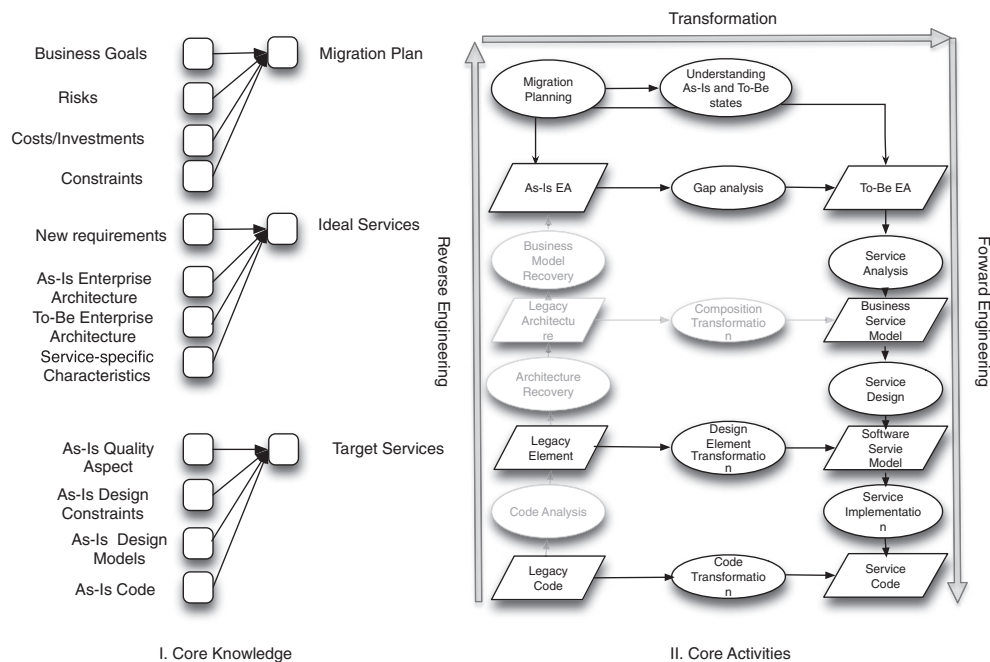


Figure 1. Lean and mean-G: general lean and mean migration strategy.

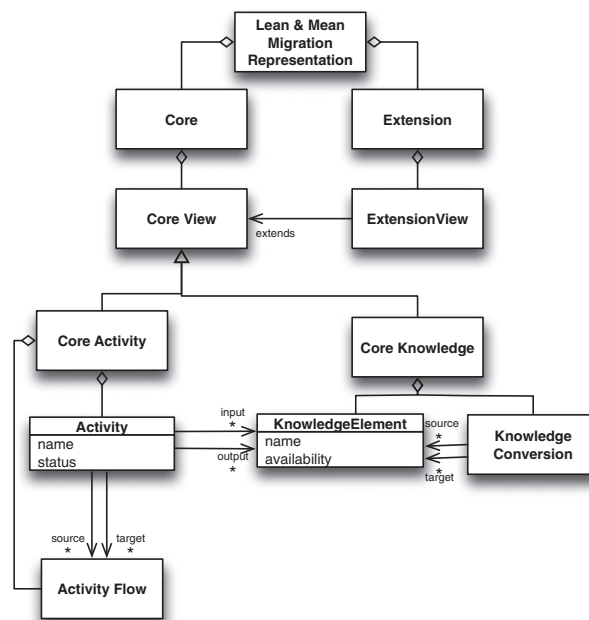


Figure 2. Extension to Core: practices bind to activities.

Core activities encompass a set of generic activities representing what needs to be done in a migration project. These activities mainly transform a number of knowledge elements. Similar to core knowledge, core activities can be customized for each migration case (Section 4.1.1 for examples).

Extension To support the more advanced needs of a certain migration project, the extensions should be appended to the core. Extensions, represented in terms of the `extension views`, are tied to the `core views` and as such enrich that view. For instance, to deal with changes incurring

during migration, an extension to knowledge view is proposed in [18]. This extension, represented in a change view, enriches the knowledge view with the concepts that are essential for identifying the impact of change. It should be noted that, customizing the core and extending the core (with an extension) are two distinct concepts. By the term customization we mean refining core elements using a number of sub-elements. For instance, in Figure 2 the knowledge element 'Data Model Complexity' is refined to two sub-elements of 'relationship types' and 'relationship cardinalities'. These sub-elements, are of the same type as 'Data Model Complexity'. By extension we mean adding a view to the core views. For instance, Figure 3 is a view which extends the activity view shown in Figure 4.

Treatment Our proposed treatment for Logica constituted (i) identifying the core and the extension(s) specific to Logica's data migration strategies (DM), and organizing it into a lean and mean strategy (referred hereafter as lean and mean-DM); (ii) for new projects, use the lean and mean-DM for defining their migration strategy based on the lean and mean-DM. Our hypothesis was that having the lean and mean-DM at hand, definition of new migration can be carried out more efficiently.

3. STUDY DESIGN

In order to define and apply the lean and mean strategies for Logica, we followed a TAR, introduced by Roel Wieringa [19]. Our primary goal for choosing TAR is in what they call practice goal in TAR, namely, to collaboratively improving a real situation and to learn from it. In the context of this work, the situation refers to efficiency problem in migration projects of Logica. In addition to our primary goal, we had a secondary one that complies with research goal in TAR. The research goal in TAR relates to increasing the relevance of research artifact (e.g., lean and mean migration strategy in this study) by trying it out in real situations (e.g., migration projects of Logica). In this work we desired to investigate whether the lean and mean-G is generalizable by applying it to migration projects of Logica. To pursue the two goals, we had to play two distinct roles: (a) designer, design a lean and mean strategy for improving problem of Logica; and (b) researcher, draw lessons learned about the lean and mean strategy.

By distinguishing between the practice and research goals as well as designer and researcher roles, TAR provides rigor in addressing both goals simultaneously. For each of the research and practice goals, a

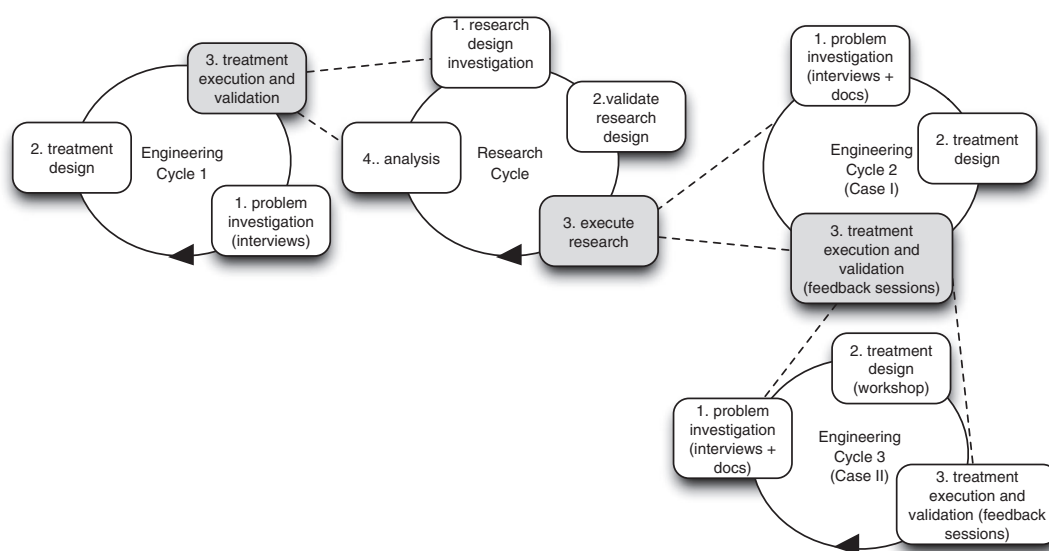


Figure 3. Research method: detailed.

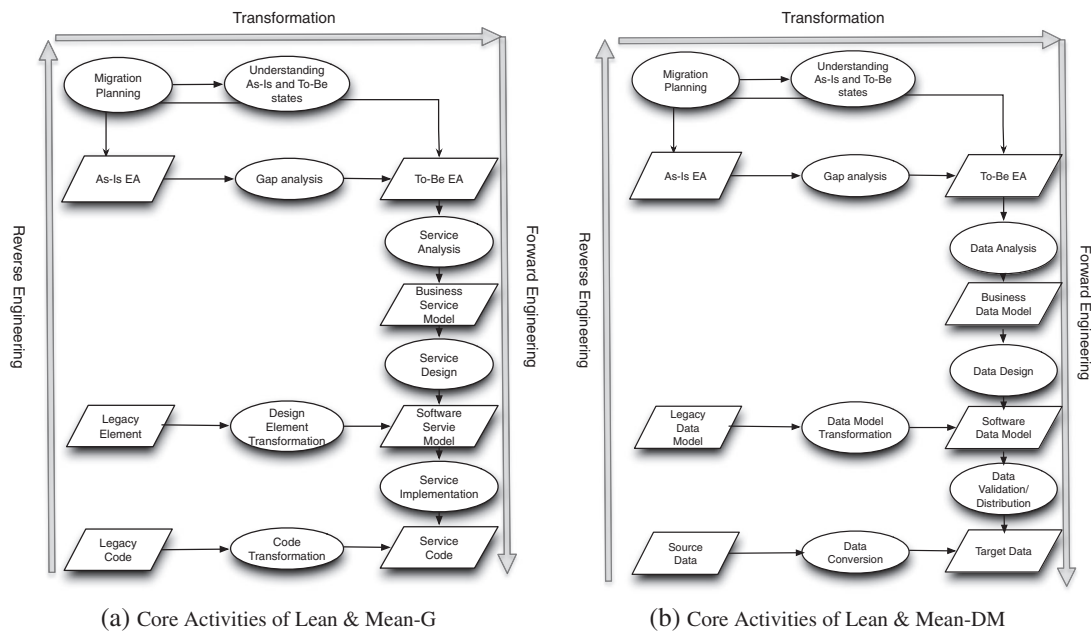


Figure 4. Customization of core knowledge for Case II.

separate action research cycle is dedicated, namely research cycle and engineering cycle, respectively. TAR is a nested, cyclical process in which each cycle could be of type research or engineering cycle. The nested structure of TAR means that there is ect, a cycle within a cycle. This is evident from Figure 5, where for instance there is a research cycle within the first engineering cycle.

The engineering cycles, addressing practice goal, include three phases: (1) problem investigation; (2) treatment design; and (3) treatment execution and evaluation. Problem investigation refers to identification of the primary problems. Treatment design refers to designing improvement solutions as treatments for the identified problems. Treatment execution and evaluation first applies the treatment and then evaluates its application. The research cycles entail the typical phases for

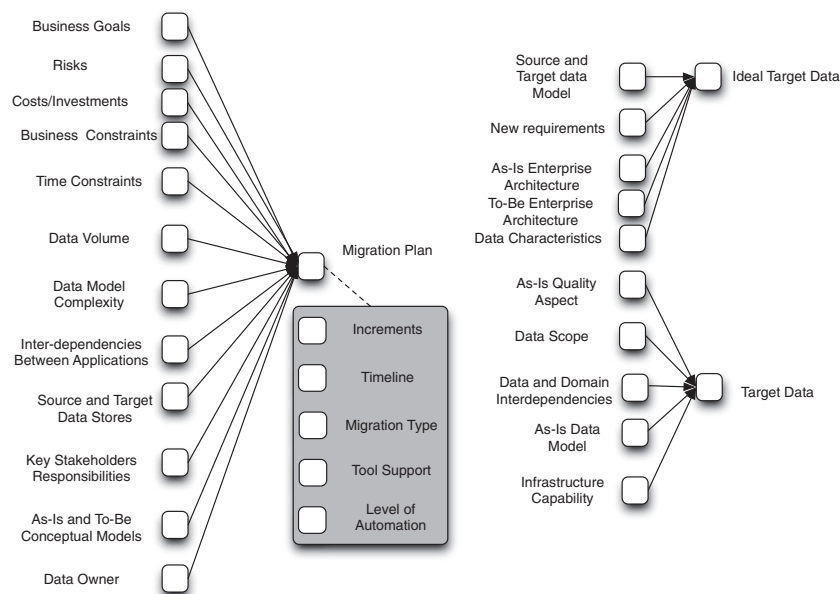


Figure 5. Research method.

research design including the following: (1) research design investigation; (2) validate research design; (3) execute research; and (4) analysis.

The TAR has been used in our industrial study as illustrated in Figure 5. As shown in this figure, we carried out three engineering cycles and one research cycle. Table II provides an anonymous profile of the participants of our study. In the following, we summarize the goals and the main results of each of these cycles. More details about the study design of the cycles can be found in Appendix A.

We carried out Engineering Cycle 1 aiming at, firstly, understanding the context and needs of Logica, and secondly, deciding on the treatment plan. These two goals were achieved following ect phase 1: problem investigation and phase 2: treatment design, respectively. In short, the treatment plan constituted first defining the lean and mean-DM based on a past project of Logica (i.e., Case I) then applying it in a new project (i.e., Case II). Phase 3: treatment execution and evaluation, initiated the research cycle.

The Research Cycle was carried out with the goal of trying out our past research artifact, lean and mean-G in real situations and learn about its uses and its generalizability.

Accordingly, this cycle addresses the following two research questions (RQ):

- RQ1. What would be the effect of having the lean and mean migration strategy in place?
- RQ2. Is the lean and mean migration strategy generalizable? By generalizability we mean if the core knowledge and core activities are common to every migration approach.

It should be noted that, RQ1 refers to our practice goal and RQ2 refers to our research goal. To answer the research questions, in phase 1: research design investigation, we designed our research (Appendix A). In phase 2: validate research design, we identified the threats to validity of this study and took actions to mitigate them (Section 8). Phase 3: research execution initiated two engineering cycles, each conducting a different case study (i.e., Case I and Case II). After completion of the last two engineering cycles in phase 4, analysis of the results was carried out. The regarding details such as unit of study and data analysis method are represented in Appendix A. The main goal of *Engineering Cycle 2* was to elicit the migration strategy of Case I and organize it into a lean and mean strategy. This cycle resulted in the first version of the customized lean and mean-G for data migration, so called lean and mean-DM. We constructed the core by (i) mapping the migration strategy of Case I on the lean and mean-G using the coding procedure presented in Section 10; and (ii) refining the mapping based on the stakeholders feedbacks received in feedback sessions. We constructed an extension to the core based on the feedbacks we received from the stakeholders as well as by analyzing the project documents. lean and mean-DM was further refined in the next cycle. In Engineering Cycle 3, we aimed at identifying the benefits of the lean and mean-DM when determining the strategy of the new project, that is, Case II. To this end, we organized a workshop with the team members. Throughout this workshop, we observed the ways in which the lean and mean-DM helped stakeholders in determining the Case II migration strategy. The observed benefits were further discussed with the stakeholders in the feedback sessions.

4. LEAN AND MEAN STRATEGY FOR DATA MIGRATION

The lean and mean-DM was constructed with the data gathered form the last two cycles of this study. Altogether, there were 208 min of feedback sessions and workshops and around 100 pages of project

Table II. Technical action research participants.

Participant	Role	Experience (years)
LNR	Sector manager	12
PTW	Chief architect	20
WDE	Business analyst	8
MAF	Architect	13
MAM	Architect	13
MTF	Project manager	5
MTM	Project manager	5

documents. The lean and mean-DM capture the kinds of activities that are fundamental in data migration (in Figure 6) and the core knowledge elements to be made available in data migration projects (in Figure 2). In addition, the lean and mean-DM also provides an extension to the core that binds domain-specific practices to the core activities (Figure 3). The practices were identified and codified through close collaboration with stakeholders. In the following, we explain the core and the extension.

4.1. Core of lean and mean data migration

This section describes the core and its organization in two views of activity and knowledge.

4.1.1. Core activities. Figure 6 shows the core activities that we found being carried out in data migration endeavors. These are the activities that are not only shared in both Case I and Case II, but are also perceived by practitioners being fundamental in DM strategies. Although the core activities of lean and mean-DM are grounded on our two case studies, they are constructed by being mapped on the core activities of lean and mean-G. In other words, core activities of lean and mean-DM are specialization of the core activities of lean and mean-G. This is evident from Figure 4 where some of the activities of lean and mean-G (Figure 4) are repeated in lean and mean-DM (e.g., migration planning or gap analysis in Figure 6), whereas some are specialized for data migration (e.g., data model transformation, instead of legacy element transformation). In the following, we briefly introduce the core activities of lean and mean-DM and our related findings.

- **Migration planning.** Migration typically starts with planning. RNL said, ‘we start planning the migration during our bidding process and at this stage we need to take a lot of important decisions about how to address the customer’s requirements while remaining in time and budget limits’. We found that the plan reflects decisions such as the number of increments, the migration timeline, toolsets, level of automation, and the level of business user involvement. Further, in the project, the plan helps the migration team to monitor the progress against the plan.
- **Understanding the As-Is and To-Be states.** Obtaining the just enough understanding of the As-Is and To-Be states was indicated as one of the key challenges of migration projects. To do this,

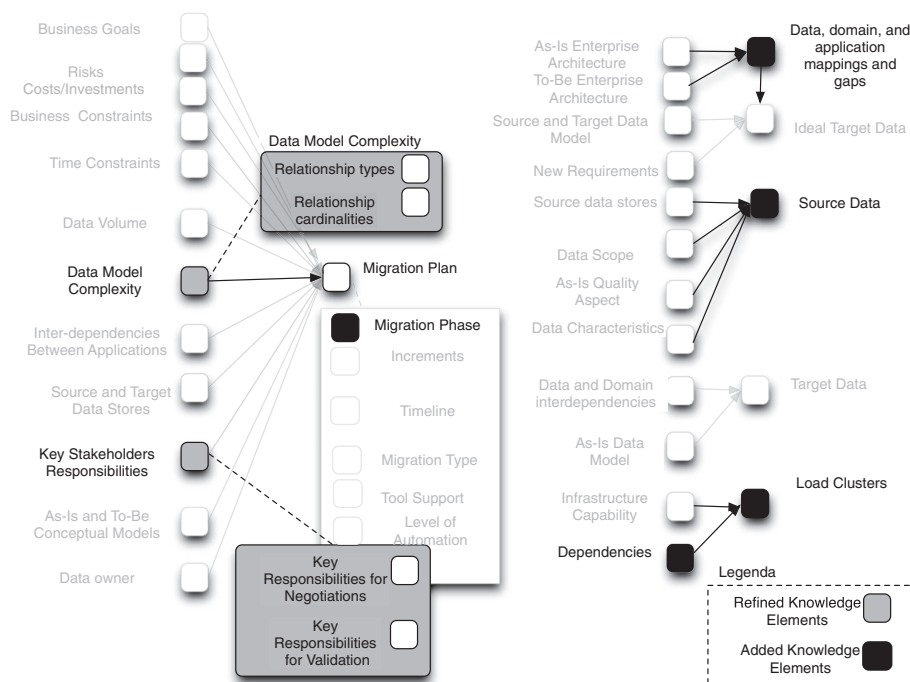


Figure 6. Core Knowledge of lean and mean data migration.

Logica extracts the relevant portion of the enterprise architecture (EA) of these two states. WDE said: 'In migration projects EA provides us the starting-point and the destination.' The As-Is and To-Be states are captured in terms of EA elements, that are, business architecture, data architecture, and technology architecture.

- *Gap Analysis.* Once the As-Is and To-Be states are understood, gap analysis at the EA level is carried out. Gap analysis aims at understanding the gaps between the business structure, data model, and technology architecture of As-Is and To-Be states. Once these gaps are identified, the migration team decides how to fill them. For instance, gaps between the conceptual data models of the two banks indicated the differences in the banking products offered by the two banks. To fill this gap, the migration team uses an internal practice, called mapping workshops, designated for negotiations and handling gaps.
- *Data Analysis.* Within this activity, the To-Be business data model is identified. We found that to carry out this activity, typical requirement engineering techniques such as conceptual data modeling are used to define the ideal data model after migration.
- *Data Design.* This activity relates to design of the To-Be data model. In doing this, the practitioner has to do trade-offs between the desired To-Be data model and the existing data model of both the source and target systems.
- *Data Model Transformation.* Within this activity, the As-Is data model is transformed to To-Be data model.
- *Data Conversion.* This activity relates to automatic conversion of the source data to the target data.
- *Data Validation and Distribution.* The practitioners emphasized that minimizing the cascading effects of rollbacks in case of a faulty data insertion is an important challenge in data migration. Data validation addresses this challenge by assuring the correctness of the data before data load. After validation, data is distributed into smaller sets to meet timeliness and consistency.

4.1.2. Core knowledge. Figure 2 captures the key knowledge elements involved in data migration projects and as such represents the essential information to work with. Each of the core knowledge elements addresses a different aspect of the migration. For instance, some knowledge address the 'why' question that underpins a project (e.g., business goals motivating the migration), whereas other knowledge addresses 'what' (e.g., source and target EA) and 'how' (e.g., emmigration type in the migration plan). Regardless of their type, these knowledge elements are the ones that repeatedly occur in data migrations. Knowing these knowledge elements, practitioners can assess early enough the costs and risks of eliciting the required knowledge and plan accordingly.

The arrows in Figure 2 reveal that to create a certain output, there are certain types of knowledge that should be available. For instance, to create the migration plan the input knowledge such as business goals, data volume, time constraints, and risks have to be made available (left part of Figure 2). This addresses a key issue emphasized by practitioners, that is, discovering what needs to be made available too late in the process [12]. The core knowledge includes three main output knowledge elements: migration plan, ideal target data, and target data. To create the output knowledge elements, altogether, 22 knowledge elements were used as input.

Core knowledge does not represent all the required information in a certain data migration project, but it only represents the bare essentials. As noted in Section 2, core knowledge can be customized to represent the key input/output knowledge elements of a certain project. Figure 7 shows an example of such customization for Case II. This customization can be realized by adding new knowledge elements to the core or refining the existing ones into more detailed elements. The black boxes in Figure 7 illustrate the added knowledge elements, and the grey ones represent the refined ones. As an example, for Case II, the migration team underlined that to make the core knowledge more expressive, a number of important output knowledge such as migration phase or source data should be added to the core. In this addition, the preexisting knowledge elements of the core (i.e., the grayed out boxes in Figure 7) act as anchors hooking the new elements to the core. In the same vein, some knowledge elements were required to be expressed in more detail and consequently were refined. As an example, the migration team emphasized that key stakeholders' responsibilities are too generic and has to be refined into two categories of key responsibilities for negotiations and key responsibilities for validation. As their main rationale for such refinement, they noted that business

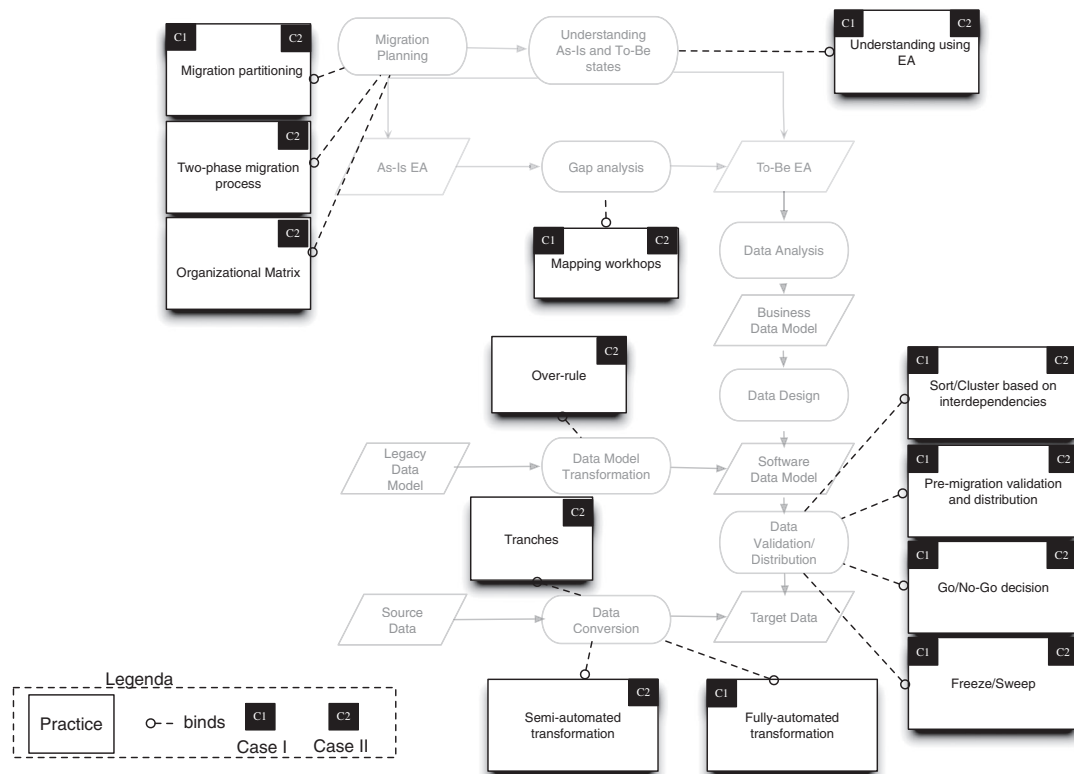


Figure 7. Extension to core: practices of related work bound to core activities.

stakeholders would take many responsibilities in the whole migration and distinguishing those responsibilities early in migration is essential. In short, based on what is fundamental in each project, the core knowledge should be enriched. In this enrichment, the core acts as an anchor to add new knowledge or to specify knowledge in more details.

4.2. Extensions

The core activities suggest ‘what to do’ without providing any specific guidance on ‘how to do it’. On one hand, the lightweight description of the activities provides directions in carrying out migration, while letting practitioners adopt their way-of-working. On the other hand, practitioners are not supported with sufficient guidance on how to do migration. This constituted one of the main topics of discussion in the feedback sessions, namely, there is a need to enrich the core activities with more guidance. This need is addressed in the extension part of the lean and mean strategy in which a set of practices are identified. We provided an extension that binds specific practices to the core activities. Figure 3 illustrates this extension where practices are shown using rectangles and are bound to the core activities using binds links. As noted, a practice provides a solution for a particular migration activity or aspects of that activity. To build the migration approach one needs, the core allows adding practices, such as mapping workshop for gap analysis, or emigration partitioning for migration planning (Figure 3). The main point here is to isolate and present practices as distinct, separate ‘solutions’ that a team can choose to use or not to use. To explicitly document the practices, we devised a template exemplified in Table III. We used the work of Clerc [20] as a basis to define this template. Using this template, one can specify a practice in terms of its main intent, the business goals that it supports, the situation in which the practice is emerged and applied, and finally the relationships between the practices. In Section 5.1.2, we will describe how elements of this template help practitioners having a well supported choice between practices for application in data migration projects.

Table III. Example of codified practice.

Best practice name	Migration partitioning
Intent	This practice intends to cope with the volume of data and/or data interdependencies by partitioning the full migration
Overview	Data migration is happening in an extended period of time (e.g., one week) instead of a point-in-time fashion. Migration is partitioned in to a number of partitions (e.g., three partitions corresponding to the three data entities including customer, product, and financial information). After each of these partitions, a baseline of data consistency is reached in the target system that allows the next partition to be inserted correctly
Business goals	Timeliness and correctness
Context	Where point-in-time migration is not possible because of extreme data volumes, complex data inter-dependencies, or other business requirements, segmentation of data into partitions is a good option. This practice can be used by freeze/sweep practice to avoid the possibility of the same piece of data being accessed and updated in both the source and target structures and systems. As such, there is no concern about collisions (concurrent updates in both locations)
Relationships	Freeze/sweep

We found 14 distinct practices, out of which seven were present in both projects analyzed in the case studies. In Figure 3, each practice is labeled with case studies that the practice was present in (*c1* for Case I and *c2* for Case II). In addition, this figure shows that some of the activities were supported by multiple practices. These practices were either complementary (i.e., they can be adopted together) or exclusive (i.e., they represent alternative ways of conducting the activity). For instance, two practices of go/no-go decision and pre-migration validation and distribution, used together in both projects, were complementary. On the contrary, semi-automated transformation and fully-automated transformation were exclusive. Regardless of being complementary or exclusive, the practices package past experience in carrying out core activities.

5. IMPLICATIONS OF LEAN AND MEAN MIGRATION STRATEGY (RESEARCH QUESTION 1)

Previous section presented the lean and mean-DM. This section discusses the effect of lean and mean-DM in defining a migration strategy of a project (RQ1). To answer RQ1 we carried out the Case II cycle (Figure 5) in which we applied the lean and mean-DM to the still active project of Case II. In the workshop of this cycle, together with practitioners (Table I), we defined the migration strategy of Case II.

The main purpose of this workshop was to investigate what the ‘real benefits’ of the lean and mean-DM are. To this aim, we observed (i) how lean and mean-DM is used in practice; and (ii) what the implications of such uses are. Our approach for conducting the first type of observations was in-line with the ‘think aloud’ approach as suggested in [21]: we steered participants to think aloud by asking them ‘How are you using the Lean & Mean-DM?’. To conduct the second type of observations, we used ‘if-then’ questions. For example, ‘If you use the core knowledge as a checklist then what the implications would be’. This way, we obtained the uses of the lean and mean-DM and their implications (columns Uses of lean and mean-DM and Implications of use in Table IV, respectively). Next, based on the observations about both *uses* and implications of use, we drew a number of conclusions. Table IV summarizes those conclusions, as well as how those were drawn. As shown in this table, some of the conclusions are directly stated by the participants, whereas others are reasoned based on the researchers’ inference by deduction. This way, we got the picture of what the benefits of lean and mean for efficiency are. These benefits fall into three main categories: (i) increased reuse; (ii) improved problem-solving; and (iii) supporting the existing way of working. These benefits are listed in three key findings presented in the following sections. Each finding is summarized in a Reflection Box, and is followed by a detailed discussion of the finding.

Table IV. Overview of findings.

Finding	Sub-finding	Uses of lean and mean-DM	Implication of use	Conclusion	Research method of conclusion drawing
F1. Increases reuse	F1.1	Core activities were used as a reminder of 'what needs to be done'	Reuse of know-how about the generic activities	Core activities facilitate reuse of know-how by acting as a reference model	Direct statement of practitioners: 'core not only externalizes the know-how, but it combines it into a coherent whole'
		Core knowledge was used as a checklist to characterize the migration process	Reuse of know-how about the knowledge that shapes and drives migration	Core knowledge facilitates reuse of know-how by characterizing the migration projects and eventually framing the class of migration approaches	Direct statement of practitioners: 'the core knowledge can help us define the prerequisites of certain migration approaches'
	F1.2	Extensions were used as a repository of concrete solutions	Reuse of concrete solutions for doing a certain activity	Extensions facilitate reuse of concrete solution	Direct statement of practitioners: 'the practices helps us to know what are the solution options and when and how to use them'
F2. Improved problem-solving	F2.1	The core knowledge was used as a starting point for identifying the problem	Improved problem understanding	Core knowledge improves problem-solving by improving problem understanding	Deductive reasoning: (A) core knowledge facilitates identifying problems and issues; (B) the more issues and premises are identified, the better problem-solving (According to Tang [22]) $\rightarrow$ core knowledge improves problem-solving
	F2.2	Core activities were used to frame their problem.	Improved problem-structuring	Core activities improve problem-solving by improving problem structuring	Deductive reasoning: (A) core activities structures the problem understanding; (B) the more structured the problem space, the more rational is the approach taken by designers According to Zannier <i>et al.</i> [23] and Schon [24]) $\rightarrow$ Core activities improves Problem-Solving
F3. Supporting existing way of working		Core was set as a reference model and was used to tailor their own company-wide approach	Supporting existing way of working by adding or removing company-wide practices	Organizing migration strategies into lean and mean supports existing way of working	Direct statement of practitioner: 'This [the lean and mean] orders what we already do and what we already know works'
DM, data migration.					

5.1. Increased reuse

Reflection Box.1. Organizing migration strategies into lean and mean increases reuse.

- **F1.1.** The core facilitates reuse of know-how.
- **F1.2.** The extension facilitates reuse of concrete solutions.

In this study, we found that organizing migration strategies into lean and mean increases reuse of migration approaches among migration projects. The direct relation between reuse of approaches and efficiency has been widely acknowledged [25, 4, 26]. According to Basili *et al.* [4], efficiency is very often achieved by reusing and modifying over and over the same approaches. This is in-line with what was perceived by practitioners in this study; by facilitating reuse of (parts of) previous migration approaches, lean and mean-DM helps decreasing the effort needed for defining migration strategies. Interestingly enough, we found that the core and the extension of lean and mean-DM contributes to two distinct types of reuse: (i) reuse of know-how; and (ii) reuse of ready-to-use solutions.

5.1.1. Core facilitates reuse of know-how (F1.1). Prior to this study, the know-how of migration (i.e., knowledge of how to do migration) remained mostly tacit in stakeholders' minds. Nevertheless, in the feedback sessions of the last cycle (Engineering Cycle 3) the migration team emphasized that the core not only externalizes such know-how, but also combines it into a coherent whole. This led us to the finding F1.1, that is, the core serves as a consolidated know-how. For instance, WDE referred to such consolidated know-how as 'a skeleton of activities and knowledge'. In this regard he said: '... it is true that there will be more or less a lot of uncertainties at this point but this core makes a good starting point for us. It shows the skeleton of activities and information'.

We further found that the way two parts of the core (core activities and core knowledge) facilitate such reuse of know-how is very different. In the following, we represent our findings in this regard.

Core activities as a reference model of migration activities During the workshop of the last cycle, we observed that the migration team took the core activities of the lean and mean-DM and built the project activities of Case II on top of those activities. In doing this, they repeatedly, (i) selected an activity; (ii) recollected their past experience related to that activity; and (iii) contextualized the activity for the specific project of Case II. This is evident in the following quote where after selecting the data conversion activity (Figure 6), MTM noted their past experience related to this activity and further discussed how it needs to be contextualized: '... for data conversion we usually use mapping tables and conversion rules, and we have tools and procedures that use them, like in Case I. But in Case II, this [mapping tables and conversion rules] will not be enough. We need a mechanism for letting business users to review and eventually override the data conversions'.

In this way, the core activities of lean and mean-DM were used as a reference model isolating the know-how related to 'what needs to be done'. The migration team used this reference model as a reminder of the generic activities that happen in data migration, the activities that most probably need to be carried out in Case II as well.

Core knowledge to characterize migration projects Core knowledge, inherently, is a repository of consolidated information that are key drivers in migration. During our workshop, we observed that practitioners reuse the knowledge elements in this repository to characterize different migration approaches. Characterization of migration projects is a prerequisite for identifying reuse opportunities [4], such characterization further practitioners to relate different projects. For instance, MAM selected data volume from the core knowledge to relate Case I and Case II. In this regard (s)he commented: 'Case I had the data volume of Case I had the data volume of nearly 1.5 million whereas Case II deals with more than 140.000. This probably means that we need fewer increments in Case II'. Likewise, RNL pointed out level of business-user involvement as a characterizing factor of Case II and said 'Case II is different from Case I because of level of business-user involvement which has to be much higher'.

The migration team further emphasized that the core knowledge enables framing the class of migration approaches suitable for certain characteristics. This enables systematic reuse of migration approaches. In this regard, WDE said: ‘the core knowledge can help us define the prerequisites of certain migration approaches. For example, approaches suitable for data migrations with a high level of business involvement, or approaches for acquisitions with a single data owner’.

5.1.2. Extension facilitates reuse of concrete solutions (F1.2)

The migration team found the extension as a repository of concrete solutions for their known problems. Each practice in the extension captures a repeatable solution for recurring problems. For instance, migration partitioning practice, represented in Table III, addresses a solution for the problem of extreme data volumes and complex interdependencies between data entities. Such solutions are emerged out of experience in various data migration projects and are likely to be reused in future projects, too. By linking these solutions to the core activities, the extension addresses a key issue emphasized by practitioners [12], that is, what are the options for carrying out a certain activity. For instance, for the validation and distribution activity, we identified four practices of dependency-based sort/cluster, premigration validation and distribution, go/no-go decision, and freeze/sweep (Figure 3). By bringing order in the practices available for validation and distribution activity, the migration team was able to readily decide on use of these practices in the Case II as well.

As a coproduct of this study, the migration team found the way practices are formed being useful for deciding ‘when’ or ‘how’ to reuse a practice. For instance, we observed that to decide on ‘when’ to reuse a solution, they referred to Intent and Context in Table III. In the same vein, the Overview of practices initiated discussions addressing how to reuse a specific solution.

5.2. IMPROVED PROBLEM-SOLVING

Reflection Box.2. Organizing migration strategies into lean and mean helps problem-structuring.

- **F2.1.** The core knowledge facilitates identifying the problems.
- **F2.2.** The core activities facilitate framing the problems.

Defining a migration strategy is a type of problem-solving process that includes recognizing problems and finding ways to devise a strategy to solve these problems. To efficiently define the ‘right’ migration strategy, practitioners need to make many well-reasoned decisions in a timely manner. As such, improving problem-solving of practitioners would positively affect the efficiency in carrying out migration. In this study, we found that organizing migration strategies into lean and mean would improve practitioner’s problem-solving.

In this study, we observed that the migration team used the core of lean and mean-DM to systematically identify the migration problems and structure them. We expect that such improvement in problem structuring would improve practitioners problem-solving as well. We further found that the core knowledge and the core activities lend themselves in such problem-solving in different manners. In the following, we will present our findings in this regard.

5.2.1. Core knowledge facilitates problem identification (F2.1)

Some designers might not explicitly identify the problems that they need to solve in migration. Instead, they think about the solutions. For instance, instead of exploring the issues that the migration faces, they might jump into fully-automated data mapping because it is perceived as a best practice. There are some negative implications with this approach. Firstly, the problems may be over simplified and not explored thoroughly. Secondly, a designer may anchor on a solution and is reluctant to shift even in view of contradictory information. This has been found in the behavior of professional designers by Tang and van Vliet [27]. One way to avoid this behavior is to use techniques or tools that support problem identification. In this study, we observed that the migration team used the core knowledge as a tool for identifying the migration problems of Case II.

In the workshop, the migration team used the core knowledge as a starting point for identifying the problems. We found that they used the core knowledge as a checklist, which helps them not to miss important information. As noted in Section 5.1.1, the migration team used the core knowledge to characterize the project of Case II. Such characterization was often followed by recognizing the peculiarities and consequent problems that the project faces. For example, after identifying confidentiality of data as an important characteristic of Case II, MAM said: ‘the level of confidentiality of data is an issue. We need to find out how to test the data because we are not allowed to check the data of VIP customers’. MTM, referring to business constraints, continued: ‘business constraints raise issues too, for example we should keep in mind that after a point-in-time there will be no acquiree bank as a legal entity. So we cannot expect any activity from an acquiree bank after that time’.

In this way, the core knowledge worked as a tool for steering problem identification. Using this tool, practitioners can reflect the potential gaps in their understanding of the problem. In this way, overlooking key issues in design could be minimized.

5.2.2. *Core activities facilitate framing the problems (F2.2)*

After identifying the generic problems, the migration team needed to decide how to approach the specific problems in carrying out the migration activities. Throughout the workshop, they alternated between raising problems and proposing solutions. The solutions were mainly of type ‘how to carry out (a part of) an activity’. For some problems, they had an immediate solution, whereas for others, they left the solution open. In doing this, inherently, they followed the famous problem solution coevolution [28]. In software engineering, it is well known that solutions are discovered and evolved on the basis of problem structuring. Software engineering design methods, such as twin peaks [29] and problem frames [30], support coevolving of problems and solutions. In this study, however, how the migration team structured the problem was new. We observed that they framed these problems using the core activities. In other words, they used activities as containers to contextualize and accommodate their problems. This is evident in the following quote where MTM frames the issue of maximizing business involvement in the scope of gap analysis activity: ‘maximizing business involvement for gap analysis means that gaps have to be identified with active involvement of business stakeholders. The issue is to involve whom. It needs to be someone who has overall view on the domain and is aware of important gaps.’

To summarize, we found that core activities enable focusing on a certain activity and identifying or translating the migration problems for that activity. This facilitates a focused design discourse that helps to formulate the problems and find solutions (e.g., practices) efficiently.

5.3. SUPPORTING EXISTING WAY-OF-WORKING

Reflection Box.3. Organizing migration strategies into lean and mean facilitates support of existing way-of-working.

In this study, we aim at efficient support of the way people work. This goal resembles what happened in general in the software process field, where they moved from processes that are strict, precise, and comprehensive (e.g., rational unified process [7], spiral model [8], structured systems analysis and design method [9]) to light-weight, agile, and highly iterative approaches (e.g., Scrum [31], Extreme Programming [32] or Lean [6]). In this emerging perspective, being efficient comes together with being simple and being compliant with existing-way-of-working of practitioners (e.g., [6, 5, 11]). More precisely, this perspective follows the idea that more elaborate, precise, and comprehensive approaches cannot support practitioners’ challenges. Instead, approaches that are reduced to absolute bare necessities and adopt existing way of working, deliver the required quality without increasing costs and efforts [6]. However, in order to achieve these benefits, it is of great importance to bring order and systemize such existing way of working. This is what our lean and mean strategy does.

The existing way of working of practitioners lies in the heart of the lean and mean strategy. The practices, captured in extension, are systematic representations of the solutions that practitioners already adopted. By providing a very lightweight core that can be extended by practices, the lean and mean strategy systemizes what practitioners already do. What was indicated as most useful was to set the core as a reference model of their migration process and tailor their migration processes using this reference. Because this reference model emerged out of existing projects, it reflected actual way of working. Furthermore, this reference model was indicated to be helpful because it fitted the social and organizational dynamics of migration teams in Logica.

6. GENERALIZABILITY OF THE LEAN AND MEAN MIGRATION STRATEGY (RESEARCH QUESTION 2)

As noted, our research goal was to try out the lean and mean-G in the real-world migration cases and see whether the core elements are generalizable. Generalizability in this study pertains to the degree to which the knowledge elements and the activities of the core in the lean and mean-G are common in the migration efforts. Lean and mean-G was the result of an industrial survey in different types of companies (in-house and consultancies) and different market positions. In the two case studies discussed in this paper, we confirmed its general applicability, at least in the specifics of data migrations. Further, generalizability issues are discussed in Section 8.

By following TAR, the lean and mean-G was changed in such a way that it matches the strategy taken in the regarding strategies of Case I and Case II. In addition, the feedback sessions resulted in further changes in order to assure that the core was fundamental in the data migration approaches. These changes resulted in deviation from the lean and mean-G. The results of the analysis revealed that almost all the changes were of type addition or refinement of the core elements. This makes lean and mean-DM being a specification of the lean and mean-DM. In the following, we will review the changes with respect to core activities and core knowledge.

6.1. Core activities

Stakeholders agreed the subprocesses from the lean and mean-G (reverse engineering, transformation, and forward engineering) should remain largely unchanged and that two activities of migration planning and understanding the As-Is and To-Be states should occur prior to these processes. Activities of forward engineering subprocess were refined into data analysis, data design, and data validation and distribution. This is to be expected given that the focus of these activities in lean and mean-G is on analysis and design of services, whereas lean and mean-DM focuses on a certain aspect of services, namely, data. In the same vein, the activities of data model transformation and data conversion are specifications of legacy element transformation and legacy translation, respectively. This confirms that the activities of lean and mean-G are core and the activities of lean and mean-DM are specific forms of the lean and mean-G.

6.2. Core knowledge

The complete list of changes made to the knowledge elements of the lean and mean-G are listed in Table V. Out of 15 knowledge elements of lean and mean-G, 7 were repeated without any changes in the lean and mean-DM, seven were refined, and one was removed.

It should be noted that the knowledge elements that were repeated in lean and mean-DM were the ones that stakeholders confirmed being fundamental in their migration projects. They further confirmed that these knowledge element are self-contained and do not need further refinements.

Some of the knowledge elements were too generic to adequately express the knowledge needed in the data migration projects. For instance, although stakeholder confirmed migration plan being fundamental output knowledge, they felt that it is too generic and it needs to be supported with more details. Accordingly, it was refined into the following set of knowledge elements:

Table V. Mapping knowledge elements of lean and mean-DM on lean and mean-DM.

	Knowledge element (lean and mean-G)	Knowledge element (lean and mean-DM)	Change
Input knowledge	Business goals	~	No change
	Risks	~	No change
	Costs/investments	~	No change
	As-Is enterprise architecture	~	No change
	To-Be enterprise architecture	~	No change
	New requirements	~	No change
	As-Is quality aspect	~	No change
	Constraints	Business constraints	Refined
		Time constraints	
		Data volume	
		Data model complexity	
		Inter-dependencies between applications	
		Source and target data stores	
	Service-specific characteristics	Data characteristics	Refined
	As-Is design models	As-Is data model	Refined
	As-Is design constraint	Data and domain interdependencies	Refined
		Infrastructure capability	
		Key stakeholders responsibilities	added
		As-Is and To-Be conceptual models	Added
		Data owner	Added
Output knowledge	As-Is code	Source and target data model	Added
	Migration plan	Data scope	Added
			Removed
		Increments	Refined
		Timeline	
		Migration types	
		Tool support	
		Level of automation	
	Ideal services	Ideal target data	Refined
	Software service	Target data	Refined

DM, data migration.

increments, timeline, migration type, level of automation, and tool support. With respect to migration type, MAF said ‘[...] what I am missing here is the type of migration. Typically, in our migration projects we have one (or a combination) of three strategies: big-bang data migration, incremental data migration, event-based data migration, and bi-directional synchronization. Based on the characteristic of the project we choose one of those. In the Case I well in advance we chose incremental-big bangs’. Other examples of the refined knowledge elements pertain to the differing focus of service migration as compared to data migration. Service migration focuses on creating services as output, whereas data migration focuses on creating data. Consequently, the two knowledge elements of protect ideal service and software service were changed to ideal target data and target data, respectively.

The stakeholders emphasized that the knowledge elements in lean and mean-G are fundamental but minimal. In the feedback sessions, as the assessments of the core knowledge proceeded, some knowledge elements were added to the core knowledge. Much knowledge elements that were previously left implicit were made explicit later by adding more elements such as key stakeholders responsibilities or data owner. For example, related to key stakeholders responsibilities MTF said: ‘When we are in the planning stage, we need to decide about to what extent stakeholders have to be involved. For instance, in Case I business stakeholders were decided to be involved mainly in the early phases of understanding the context, as well as in the validation. But in Case II we need much more involvement. We need to plan this in early stages of the project’.

Only one of the input knowledge elements of lean and mean-G (As-Is code) was removed. This was because the stakeholders found As-Is code of being out of scope for data migration projects. In summary, we can conclude that the core knowledge of lean and mean-G is generalizable.

7. RELATED WORKS

The work presented in this paper is related to two areas of research: lean software processes and data migration approaches. In the following, we discuss the related work in each of these two areas.

7.1. *Lean software processes*

The term ‘lean’ in software development refers to a synthesis of a system of practices, principles, and philosophy for building software systems for a customers use [33]. In their book, *Lean Software Development* [34], Poppendieck *et al.* emphasized seven principles of lean software development: optimize the whole, eliminate waste, build quality in, learn constantly, deliver fast, engage everyone, and keep getting better. Many works have applied lean principles in proposing software development processes [5, 10, 11]. Most related work is, however, Software engineering method and theory (SEMAT), which has the very broad focus of software engineering in general. SEMAT is an initiative that was introduced at the end of 2009 by Ivar Jacobson, Bertrand Meyer, and Richard Soley. This initiative recognizes the important issues being in the field of software engineering such as the abundance of unique methods that are hard to compare, and the gap between academic research and its practical application in industry. To solve these issues, SEMAT envisions a solution, which is based on the idea of separating the essence of software engineering from complex context-dependent details. The essence is captured in SEMAT kernel and the context-dependent details are represented in practices. Our approach is similar. We worked together with practitioners to identify what is the essence of data migration and represented it in terms of core. In addition, we developed the extension that brings order in the data migration practices and provides a common ground for comparing and defining practices. In this way, we have addressed the earlier mentioned issue of abundance of unique, incomparable methods. In addition, by following TAR, we took a step towards filling the theory and practice gap.

7.2. *Data migration*

Data migration approaches have been extensively investigated and discussed in the literature. Many approaches provide a general overview of the data migration approaches [35–37], whereas others [38–40] only focus on certain techniques or practices to carry out certain activities. Inherently, the first group lends itself to our concept of core, as these groups of approaches envision to be context-generic and applicable to most of data migration projects. Likewise, by proposing a context-specific and concrete way of carrying out an activity, the second group closely relates to our extensions. To compare our work with the first group, we map those generic approaches on the core of lean and mean-DM. To compare our work with the second group, we identify the distinct practices proposed in the literature and we represent them as an extension to the core.

7.2.1. *Generic data migration approaches.* The mapping of the generic approaches [35–37] on the core revealed three findings: (i) core knowledge is not explicitly covered by any of the existing approaches; (ii) in fundamental activities such as forward engineering, the To-Be data is almost neglected in the literature; and (iii) although the activities in these approaches are manifested as being common to all data migration projects, some remain quite detailed and context-dependent. Table B.1 summarizes the mappings of the generic approaches on the core activities.

Haller *et al.* [35] proposed a generic process model of the data migration projects. This work takes a project management perspective and focuses on the main activities and deliverables of the data migration projects. Activities related to reverse engineering of data, however, constitute the main focus of this work whereas forward engineering activities are neglected. Besides, although these activities are manifested as being common to all data migration projects, some remain quite detailed and context-specific. For instance, final rehearsal is a practice that might be carried out in migration projects, but it is not fundamental. With the main focus on technical aspects of data reverse engineering, the butterfly approach [36] provides a generic six-phase approach to data migration. As shown in Table B.1, this approach leaves out fundamental activities such as gap analysis, and analysis and design of target data. In the same vein, the approach presented in [37] only focuses on

In summary, the core of lean and mean-DM represent the most complete picture of ‘knowledge to be used or produced’ and the ‘activities to be carried out’ when performing data migrations.

Practices for migration planning Related to migration increments, the many practices presented in [36,41,42] fall under two main categories: (i) sudden migration (i.e., big bang and cold turkey) that signifies the case where data is suddenly and completely migrated at one moment in time; or (ii) incremental (i.e., gradual, partial, or chicken little) that relates to practices in which parts of the data are migrated incrementally. Related to the way physical data is migrated, we found two practices (i) direct migration, where the data is directly transferred from source to target; and (ii) indirect migration, where the data is extracted to a platform and is converted and tested on that platform before being loaded on the target environment.

[illegible]

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the basis of their business knowledge and experience. Manual mappings are used for migration cases where, for instance, the same data (e.g., account) is described differently in the source and target application. This was the case for the Case II; (ii) in automatic mapping the discrepancies between data models are identified automatically (e.g., [39]).

Practices for Data Conversion There are three main practices for data conversion in the literature: (i) source data inspection, where before converting data is analyzed to check the quality and structure on data instances level [35,42]; (ii) data cleans practice aims at correcting corrupted data leading to an improved data quality [43]; and (iii) data enrichment practice enriches the source data with new data before converting to the target data [35].

Practices for data validation/distribution The data validation practices proposed in [35,38] are (i) automated validation by automatic comparison of source and target data it ensures the correctness, completeness, and consistency on a structure and data level; (ii) manual validation, however, compares the source and target data based on its business domain and experience.

In summary, the classification of the practices using the core activities (exemplified in Figure 8) can act as (the first version of) a frame of reference for data migration practices. Having such a frame of reference, practitioners can make their own data migration approach using the ‘best-of-breed’ practices possibly originated in different academic data migration approaches. This contrasts with traditional approaches that treat data migration as a collection of indistinguishable practices and lead teams to abandon good with the bad when they move from one approach to another.

8. THREATS TO VALIDITY

Lincoln and Guba [44] argue that qualitative research should be, objective, reliable, and internally and externally valid. In the succeeding texts, we discuss the validity threats of this work and how we addressed them.

8.1. Objectivity

Objectivity ensures that the conclusions depend on the ‘subjects and the conditions of the case’ rather than on the researcher [45]. The emphasis here is on the replicability of a study by other researchers. A possible threat to objectivity relates to the biases stemming from researcher effects on the case. To mitigate this threat, we followed the suggestion of Miles and Huberman [45] and adopted the following tactics to minimize the researcher biases: (i) one of the researchers spent three days a week full-time for a period of two months on-site. This helped us to be less of an outsider and minimize biases leading to practitioners crafting their responses to show a more positive picture of ‘the way migration is carried out’; (ii) some of the interviews were off-site (either in our university or in an informal setting). This helped us to take a lower profile as well as collecting background information when we were off-site. A threat to replicability of results is that the record of study’s methods and procedures is not explicit or detailed enough. To address this threat, besides explaining our study design in detail (Section A.1), we explicitly linked the conclusions with the observations and the methods of conclusion drawing (Table IV). This helps other researchers to follow actual sequence of how data was collected and processed for our conclusion drawings.

8.2. Reliability

The underlying issue of reliability is whether the process of study is consistent over time across researchers and methods [45]. To ensure the reliability of our study, the study design (shown in Figure 9) was reviewed by an expert in the field of empirical software engineering. In addition, the results of coding, conducted by the first author, were peer-reviewed by the second author. These tactics helped us ensure that the study is done with reasonable quality.

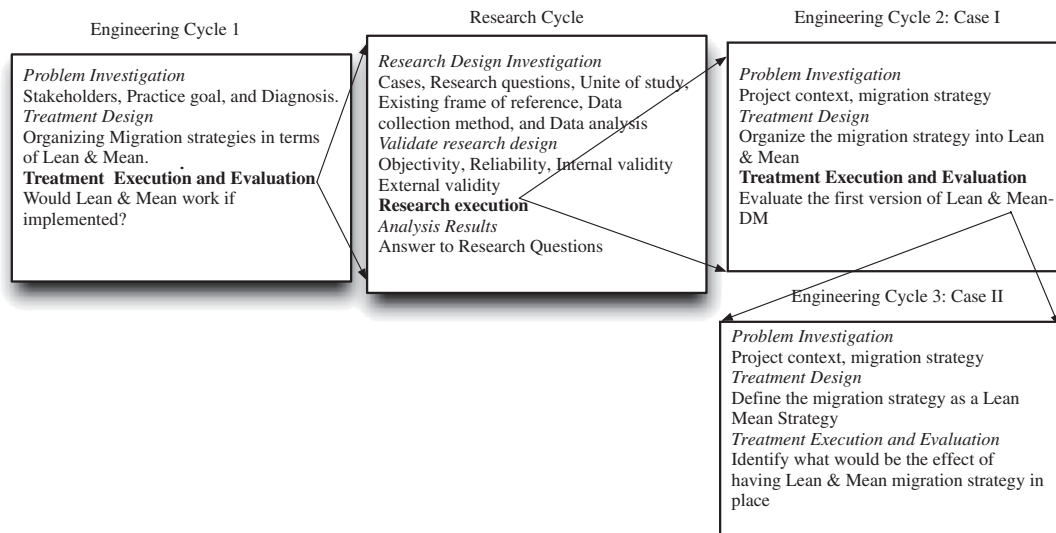


Figure 9. Meta-model of the lean and mean migration strategy.

8.3. Internal validity

Internal validity aims at ensuring that the collected data enables researchers to draw valid conclusions [46]. This validity relates to 'how' the research is carried out and whether the used methods are credible. Yin [47] suggests triangulation as tool for ensuring internal validity. The basic idea is to gather different types of evidence to support conclusions. Internal validity of this study is supported by triangulation at data source and method level. Data source triangulation was achieved because data (e.g., activities carried in migration) were gathered multiple times from documents, interviews, and observations in the feedback sessions and workshops. Method triangulation was achieved by combining different types of data gathering techniques: interviews and observations and archival data.

In this study, the interviews were mainly conducted by a single researcher and hence subjective interpretations might exist. To mitigate this threat, the interview guide was checked and validated by senior researchers experienced in software engineering, empirical studies, and migration. The role distribution of participants in this study in Table II shows that a population selection bias based on role is unlikely. We also intentionally chose senior practitioners with considerable experience in migration projects. This gives confidence that participants are familiar with migration projects throughout Logica.

Threat to the validity of the analysis is in general applicability of the codes used for characterizing and classifying migration approaches. An assuring factor in this regard is that the start-list of codes is extracted from the lean and mean-G published in a service-oriented computing forum, after being peer reviewed by experts in the field [12]. In addition, the lean and mean-G stems from an interview with seven SOA solution provider companies. This further consolidates its general applicability. An additional threat is that the analysis is performed by a single researcher. Nevertheless, bias and omission issues are (partially) mitigated by having the codification results double-checked by both a second researcher and the interviewees. Finally, to assure accuracy of findings, we used the member checking method, that is, getting feedback on the findings from the subjects who provided the data in the first place. Member checking was extensively used as the lean and mean-DM and initial findings were refined and confirmed in the feedback sessions of the last two engineering cycles. In addition, adopting TAR with the main focus on the practitioner's point of view on usability and effects provided a sound correction of the bias of the researchers.

8.4. External validity

External validity defines to what extent findings from the study can be generalized [46]. In this regard, a possible threat, inherent in any study based on case study research, is that the results are subjective.

Yin points out that external validity can be improved by using replicated study designs [47]. In this study report, we present results that are based on findings made in two different case studies. This reduces the influence of the concrete cases. A threat to generalizability of results is that the study was conducted at one company that means the findings are specific to this study. In order to cover different but relevant perspectives on the subject matter, we chose practitioners with different roles of chief architect, sector manager, business analyst, and project manager. In addition, we chose senior stakeholders who hold extensive experience and are aware of the characteristics of migration projects. This paves the way in favor of generalizability of our results.

9. DISCUSSION AND CONCLUSION

This paper presents a lean and mean strategy for data migration experimented in technical action research in industry. The research presented here has addressed a novel approach for organizing migration strategies called the lean and mean strategy. This approach is based on the idea of separating the core from the extensions. To the best of our knowledge, this is the first study of the kind in the context of migration.

Of course, various studies present approaches for legacy migration [2]. All these approaches aim at being adopted in practice. However, results show [15] that most academic migration approaches are not applied in practice as is because they do not fit their problems and needs. In this work, by actively involving practitioners, we were able to identify the important problems of practitioners as well as to provide a sound base for the lean and mean strategies.

The industrial perspective of this study brought interesting observations and lessons learned and as such contributed to the novelty and industrial relevance of our approach. The reactions to the lean and mean strategy led to some usage implications that we, as researchers, did not expect them. In addition to the effects discussed in Section 5, we identified the following lessons learned:

- *Core is the starting point.* During this study, practitioners repeatedly emphasized that although each project is unique, and the migration processes used in them cannot be simply repeated, starting from core can save them major efforts. They confirmed that core is generic enough to have wide applicability and as such, it helps them avoid starting from scratch.
- *Lean and mean supports creativity.* The practitioners acknowledged that by bringing structure on relevant knowledge and activities, lean and mean should help not overlooking important information when defining migration strategies. In addition, by providing only fundamental information, and not providing a prescriptive, step-by-step procedure, lean and mean offers room for being creative.
- *Lean and mean supports teamwork.* Having a shared mental model of the approach for carrying out migration was emphasized to be crucial for successful teamwork. Practitioners acknowledged that the lean and mean strategy can act as such a shared mental model and as such it encourages teamwork.
- *Lean and mean has to be under constant evolution.* During our extensive discussion on the uses of lean and mean, practitioners emphasized that both the core and the extensions need to be under constant evolution. For instance, practices that become prevalent in most migration strategies should be moved to the core. On the other hand, the extensions need to evolve to reflect the evolution of the teams way of working. Therefore, lean and mean strategy should be adapted and improved over time by the people and teams doing the work.
- *Company-wide lean and mean in different levels of abstraction.* What was indicated as very useful by practitioners was defining and developing the core in different levels of abstraction. They emphasized that for companies like Logica, whose core business is migration consultancy, it is valuable to have a generic core for migration strategies in general, besides more specific ones for the underlying areas (e.g., application migration, data migration, product migration, etc.). In this way, the core consolidates and brings order to company-wide knowledge, practices, and policies.

- *Common ground for communication and discussion.* Having a common ground, junior and senior practitioners can better communicate. Although senior practitioners, out of experience, 'know' what needs to be done, by externalizing the core and extensions, they can more effectively communicate with and instruct the juniors. This way, seniors do not have to update and explain an overly complex methodology; they can easily describe and position their ideas on the core in a concise and reusable way. On the other hand, juniors benefit from a consolidated company-wide knowledge and experience captured in both the core and extensions.
- *Lean and mean has to be defined incrementally.* The practitioners emphasized that the lean and mean strategy should be defined incrementally. The idea is that practitioners may explore, understand, and define the activities to be carried out and the knowledge to be used incrementally as in different stages of the project; they would focus on a subset of activities and knowledge. This reduces barriers to the adoption of lean and mean by making it possible to gain value without needing to make a high investment in defining the complete strategy at once.

The essence of our approach are the following: the core elements that are repeating in different projects must be separated from the project-specific ones; the advanced needs of certain projects should be supported as an extension to the core; and the core and the extensions should be under constant evolution. Making the fundamental elements visible provides the necessary information for making informed decisions about what is expensive to change, such as what should drive the migration (business goals or To-Be architecture) or what activities must be performed.

This study discussed how organizing migration strategies in form of lean and mean positively affects efficiency of migration. This result is based on the experience with the effects of lean and mean-DM in Case II. Although initial results are promising, future work includes empirical validation of the effects of lean and mean in efficiency to be measured in terms of cost to value ratio. This requires engagement of a broader set of projects.

APPENDIX A

Study design-appendix

In this section, our study design is described in detail.

ENGINEERING CYCLE 1

Problem investigation

In this phase, we came to an understanding of the stakeholder participating in our study, context, and needs of Logica through a number of interviews. Interviews were used in order to collect qualitative data about the context, goals, and problems diagnosis. Three stakeholders involved in different aspects of migration were interviewed: the chief architect, a business analyst, and a sector manager. In order to remain as open as possible in identifying practitioners problems, we chose open interviews.

- *Stakeholders:* The stakeholders interviewed for investigating the problem include the following: involved in different aspects of migration were interviewed (i) the chief architect with more than 20 years in migration projects; (ii) the business analyst with 8 years of experience; and (iii) the sector manager with 12 years of experience.
- *Practice goal:* Increasing the efficiency in the way migrations are carried out.
- *Diagnosis:* Inadequate reuse of the past experience in their migration projects.

Treatment design

In this phase, we decided on the treatment plan. The plan was to introduce lean and mean strategies to Logica migration strategy and first recover the lean and mean strategy of a past project and later use

this strategy in an ongoing project. Accordingly, together with the stakeholders, we chose one past and one ongoing data migration project. These two projects are introduced in Section 1.1.

- *expected effects*: (i) improvements in reusing past experience; (ii) reduction of staff efforts by being lightweight and flexible.

Treatment execution and evaluation

Within this phase we executed the plan. This phase initiated the research cycle. The execution of the lean and mean resulted in

- *effects of lean and mean in the context*: the uses of lean and mean strategy reveal the following benefits each related to efficiency: (i) increased reuse (ii) improved problem-solving (iii) capturing and supporting existing-way-of-working (Section 5).

RESEARCH CYCLE 1

Research design investigation

In this phase, we defined the cases, research questions, unit of study, existing frame of reference, data collection method, and data analysis method.

Research questions:

- RQ1. What would be the effect of having the lean and mean migration strategy in place?
- RQ2. Is the lean and mean migration strategy generalizable?

Case selection

In order to answer to RQ1, we needed to obtain the lean and mean strategy of a past project. Having such lean and mean strategy at hand, then we could observe its effects on defining the migration strategy of a new project. Accordingly, we needed a past project (i.e., Case I) and an ongoing project (i.e., Case II).

Unit of study

The data population of this study constituted the migration projects in the Logica.

This study is an embedded case study with two different projects as the unit of analysis. The unit of analysis was selected to fit the specific research and practice purpose.

Existing frame of reference

In this study, no explicit theories are referred to. However, the way migration strategies are identified is based on existing lean and mean-G. Earlier studies thereby affected the design of this study.

Data collection method

Data (i.e., qualitative data) were primarily collected using interviews, archival data, and observations. The interviews conducted were semi-structured with open-ended questions, which allow interviewers to ask follow-up questions when necessary. This flexibility allows the interviewer to ask the interviewees detailed questions about their specific areas of expertise. The interviews mainly helped us to understand the context and the taken/desired migration strategy.

In our study, archival data was a complementary source of information. In Case I, the project was conducted prior to study, which meant that the data from this project was analyzed in retrospect.

The observations were conducted in the workshop of Case II to observe the effects of having lean and mean-DM in place. In these observational studies, we steered participants to think aloud (Section 5). Considering the categories of observational studies proposed in [48], our study is of type ‘High degree of interaction with researcher’ (because we were actively participating in the feedback sessions/workshops) and ‘High awareness of being observed’. The data collected consisted of video recording of feedback sessions/workshops.

Data analysis method

The data analysis method in this study is qualitative. We chose coding as our qualitative analysis method, and we used it for two distinct purposes (i) to characterize the migration strategy of Case I and Case II and to generalize the lean and mean-DM; and (ii) to identify the effects of organizing migration strategies into lean and mean. In the following, we describe our coding in approach related to each of the two purposes.

To typify the migration approaches, we codified the interviews, and the feedback sessions. The analysis resulted in the characterization of the migration approaches on the basis of activities carried out and the knowledge used and produced. To carry out the analysis systematically, we devised a coding procedure. For creating codes, we followed the suggestion of Miles and Huberman [45] and started with an initial set of codes called ‘start-list’ and we refined it during the analysis. Our start-list stems from the lean and mean-G introduced in Section 2. The coding procedure is described in the following:

1. *Filling in activities involved in migration*: coding the activities involved in the migration. Filling in refers to creating new codes as insights occur during the analysis.
2. *Surfacing knowledge elements*: identifying the input/output knowledge elements of migration activities as well as the conversions among them. This way, the knowledge elements that are used and/or produced throughout the migration approach are extracted.

To identify the effects of lean and mean on efficiency, we used coding, too. This time, our start-list included two codes only: uses-of-lean and mean and usage-implications. Our coding procedure was surfacing uses and usage implications. This way, we extracted the way lean and mean is used as well as what are the implications of such usage.

Validate research design

The research design was evaluated to get an impression of its objectivity, reliability, and external and internal validity. The details are discussed in Section 8.

Research execution

This phase initiated the next engineering cycles each conducting a different case study (i.e., Case I and Case II).

Analysis results

Using the coding procedure introduced earlier in the text, we mapped each of the migration strategies of Case I and Case II on lean and mean-G. We further analyzed the feedback sessions and the workshop. The analysis resulted in answering our research questions.

ENGINEERING CYCLE 2

The main goal of this cycle was to recover the migration strategy of Case I and organize it into a lean and mean strategy.

Problem investigation

The main goal of this phase was to understand the context of the project through interviews and existing documents. For interviews, three stakeholders who had holistic view on the project were selected: the chief architect, the project manager, and a business analyst.

Treatment design.

Using our coding procedure, we first mapped the strategy of Case I on the lean and mean-G.

Treatment execution and evaluation

At this phase, the lean and mean strategy of Case I was evaluated within the feedback sessions. During the feedback sessions, each of the stakeholders was asked to identify knowledge or activities that are core and are missing in the mapping. For example, for each of the activities they were asked:

- Should this activity occur in all data migration projects?
- What knowledge is used in or produced by this activity?

The purposes of feedback sessions were twofold: (i) evaluate whether the core is really prevalent in data migration projects; and (ii) identify the special needs of Logica that pinpoint the needed extensions to the core. In the feedback sessions, we observed and asked exploratory questions to provoke reactions that differed from our expectations. The feedback sessions resulted in the first version of lean and mean strategy of data migration projects called lean and mean-DM. The lean and mean-DM was further challenged and refined in the engineering cycle 3. Section 4 describes the lean and mean-DM.

Engineering Cycle 3. Our goal in the third engineering cycle was to identify the benefits of having the core and extensions of the lean and mean-DM when determining the strategy of the new project, that is, Case II.

Problem investigation

The first stage in this cycle constituted understanding the context of Case II. Similar to Case I, such understanding was gained through interviews and existing documents.

Treatment design

At this phase, to determine the migration strategy for Case II, we organized workshops with project stakeholders: the project manager, the chief architect, and a business analyst. These workshops resulted in defining the lean and mean strategy of the project. Throughout these workshops, we observed the ways in which the lean and mean-DM helped stakeholders in determining the Case II's migration strategy.

Treatment execution and evaluation

The observed benefits were further discussed with the stakeholders in the feedback sessions. The lean and mean-DM was further revised based on the stakeholders' feedback.

APPENDIX B

Overview of mapping the related works on core activities

Table B.1 provides the mapping of the generic migration approaches proposed in literature on the core activities.

Migration approach	Migration planning	Understanding As-Is and To-Be states	Gap analysis	Data analysis	Data Design	Data validation/distribution	Data model transformation	Data conversion
[35]	Call for tender and bidding, strategy pre-analysis	Source and target structure and data analysis	N/a	N/a	N/a	Tests: •run •appearance •completeness & type correspondence •process-ability •integration •finalizing	N/a	Data transformation
[36]	Migration decision making	Understanding: •the preliminary requirements •the target architecture •the target hardware system •the legacy data •the to-be-migrated data •the legacy applications •to-be-migrated data	Identifying differences between As-Is and To-Be data	N/a	N/a	Freeze and cut-over	N/a	Transform the legacy data into the target data
[37]	N/a	N/a	N/a	N/a	N/a	N/a	Wrap data access	Translate data to native format and indexed files to database tables, normalize the data design

N/a, not applicable

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REFERENCES

1. Gu Q, Lago P. Exploring service-oriented system engineering challenges: a systematic literature review. *Service Oriented Computing and Applications* 2009; **3**(3):171–188.
2. Razavian M, Lago P. A Frame of Reference for SOA Migration. Towards a Service-Based Internet. Springer: Berlin/Heidelberg, 2010; 150–162.
3. Hefley B, Murphy W (eds). Service Science, Management and Engineering Education for the 21st Century. Service Science: Research and Innovations in the Service Economy. Springer US: Boston, MA, 2008.
4. Basili VR, Caldiera G, Rombach HD. The experience factory: Packaging Software Experience. Encyclopedia of Software Engineering vol. 2 (1st edn). Marciniak JJ (ed.). John Wiley & Sons, Inc.: New Jersey, USA, 1994; 469–476.
5. Johnson P, Ekstedt M, Jacobson I. Where's The theory for software engineering? *IEEE Software* 2012; **29**(5):96–96.
6. Ebert C, Abrahamsson P, Oza N. Lean software development. *IEEE Software* 2012; **29**(5):22–25.
7. Jacobson I, Booch G, Rumbaugh J. *The Unified Software Development Process, Object Technology* vol. 37. Addison-Wesley: UK, 1999.
8. Boehm B. A spiral model of software development and enhancement. *ACM SIGSOFT Software Engineering Notes* 1986; **11**(4):22–42.
9. Rose G. SSADM-the open methodology. *IEE Colloquium on an Introduction to Software Design Methodologies*, IET, 1992; 6/1–6/5.
10. Nord RL, Ozkaya I, Sangwan RS. Making architecture visible to improve flow management in lean software development. *IEEE Software Sep* 2012; **29**(5):33–39.
11. Kruchten P. A plea for lean software process models. *Proceeding of the 2nd workshop on Software engineering for sensor network applications - SESENA '11*, ACM Press, 2011; 235.
12. Razavian M, Lago P. A lean and mean strategy for migration to services. *WICSA/ECSA 2012 Proceedings Companion Volume*. ACM, 2012; 61–68.
13. Friedman T. Risks and challenges in data migrations and conversions. *Technical Report*, Gartner Inc. 2009.
14. Wieringa R, Morali A. Technical action research as a validation. *7th International Conference, DESRIST 2012*, Springer Berlin Heidelberg, 2012; 220–238.
15. Razavian M, Lago P. A Survey of SOA Migration in Industry. International Conference on Service Oriented Computing, ICSOC, vol. 215483. Springer Berlin: Heidelberg, 2011; 618–626.
16. Losavio F, Chirinos L, Matteo A. ISO quality standards for measuring architectures. *Journal of Systems and Software* 2004; **72**(2):209–223.
17. van Vliet H. Software Engineering: Principles and Practice (3rd edn). John Wiley & Sons: New Jersey, USA, 2008.
18. Razavian M, Lago P. A viewpoint for dealing with change in migration to services. *2012 Joint Working Conference on Software Architecture & 6th European Conference on Software Architecture*. IEEE, 2012; 201–205.
19. Wieringa R, Moral A. *Design Science Research in Information Systems. Advances in Theory and Practice, Lecture Notes in Computer Science*, vol. 7286. Springer Berlin Heidelberg: Berlin, Heidelberg, 2012.
20. Clerc V. Towards architectural knowledge management practices for global software development. *Proceedings of the 3rd international workshop on Sharing and reusing architectural knowledge - SHARK '08*, ACM, 2008; 23–28.
21. Owen S, Brereton P, Budgen D. Protocol analysis. *Communications of the ACM Feb* 2006; **49**(2):117–122.
22. Tang A. Software designers, Are You biased ? *Sixth workshop on SHaring and reusing architectural knowledge (SHARK 2011)*, ACM Press, 2011; 1–8.
23. Zannier C, Chiasson M, Maurer F. A model of design decision making based on empirical results of interviews with software designers. *Information and Software Technology* 2007; **49**(6):637–653.
24. Schön DA. The Reflective Practitioner: How Professionals Think in Action. Basic Books, 1991.
25. Boehm B. Managing software productivity and reuse. *Computer* 1999; **32**(9):111–113.
26. Henninger S. An evolutionary approach to constructing effective software reuse repositories. *ACM Transactions on Software Engineering and Methodology (TOSEM)* 1997; **6**(2):111–140.
27. Tang A, van Vliet H. Design strategy and software design effectiveness. *IEEE Software* 2012; **10**(1):61–63.
28. Dorst K, Cross N. Creativity in the design process: co-evolution of problem solution. *Design Studies* 2001; **22**(5):425–437.
29. Nuseibeh B. Weaving together requirements and architectures. *Computer* 2001; **34**(3):115–119.
30. Jackson M. Problem frames and software engineering. *Information and Software Technology* 2005; **47**(14):903–912.
31. Schwaber K, Beedle M. *Agile software development with scrum, series in agile software development*, vol. 18. Prentice Hall, 2001.
32. Beck K. Extreme Programming Explained: Embrace Change (2nd edn). Addison-Wesley Professional: Boston, 2005.

33. Poppendieck M, Cusumano MA. Lean software development: a tutorial. *IEEE Software* 2012; (September/October 2012):26–32.
34. Poppendieck M, Poppendieck T. Lean Software Development: An Agile Toolkit. Addison-Wesley Professional: New Jersey, USA, 2003.
35. Haller K, Matthes F, Schulz C. A Detailed Process Model for Large Scale Data Migration Projects. *Business Information Systems, Lecture Notes in Business Information Processing*, vol. 117. Abramowicz W, Kriksciuniene D, Sakalauskas V (eds.). Springer Berlin Heidelberg: Berlin, Heidelberg, 2012; 165–176.
36. Wu B, Lawless D, Bisbal J, Richardson R, Grimson J, Wade V, O'Sullivan D. The butterfly methodology: a gateway-free approach for migrating legacy information systems 1997; 200.
37. Rahgozar M, Oroumchian F. An effective strategy for legacy systems evolution. *Journal of Software Maintenance and Evolution: Research and Practice* Sep 2003; **15**(5):325–344.
38. Matthes F, Schulz C, Haller K. Testing quality assurance in data migration projects. *2011 27th IEEE International Conference on Software Maintenance ICSM*, IEEE, 2011; 438–447.
39. Ceccato M, Dean TR, Tonella P, Marchignoli D. Migrating legacy data structures based on variable overlay to Java. *Journal of Software Maintenance and Evolution: Research and Practice* 2010; **22**(3):211–237.
40. Moes T, Wijbenga J. A validatable legacy database migration using ORM. *On the Move to Meaningful Internet Systems: OTM 2012 Workshops*, Springer Berlin Heidelberg, 2012; 409–418.
41. Bisbal J, Lawless D, Grimson J. Legacy information systems: issues and directions. *IEEE Software* 1999; **16**(5):103–111.
42. Morris J. Practical Data Migration. British Informatics Society Limited: Swindon, UK, 2006.
43. Müller H, Freytag J. Problems, methods, and challenges in comprehensive data cleansing. Professoren des Inst. Für Informatik, 2005.
44. Lincoln YS, Guba EG. Naturalistic Inquiry, vol. 75. Sage Publications: New York, USA, 1985.
45. Miles MB, Huberman M. Qualitative Data Analysis: An Expanded Sourcebook (2nd edn). Sage Publications: New York, USA, 1994.
46. Creswell JW. Research Design: Qualitative, Quantitative, and Mixed Method approaches. Sage Publications: New York, USA, 2003.
47. Yin RK. Case Study Research: Design and Methods (4 edn). Sage Publications, 2008.
48. Runeson P, Höst M. Guidelines for conducting and reporting case study research in software engineering. *Empirical Software Engineering* 2008; **14**(2):131–164.

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